FLOSSI0ULLK Meta-Instruction Iteration v2.1
Unified Living Context Daemon + ROI/Leverage Substrate
(Synthesizing Project-Spine v0.5, Seed Packet v1.0.0, Context Compression v1.1, CLAUDE.md, oh-my-meta.md, FLOSSI0ULLK_Context_Daemon_Architecture.md, HI_ROI_NAO.md, Automated Agent Orchestration report, Latent Actions paper, Division of Labor paper, Cultural Semiotic Transmission paper, information theory paper, network alignment paper, stewardship governance paper, and all supporting artifacts)

Multi-Lens Snapshot (4 Lenses on the Complete Corpus)

Practical/Engineering
The full corpus now supplies a complete, battle-tested substrate ready for immediate execution. Spine v0.5 + Seed Packet v1.0.0 define the DNA (Carrier Equivalence, Two-Plane Boundary, SDD, Voluntary Convergence, Provenance). The Context Daemon Architecture + HI_ROI_NAO.md deliver the missing operational layer: a 5-layer projection service (Observer → Semantic Index → Sourcechain Projection Graph → CRDT State → Stakes-aware Curator) that eliminates 13-month context loss while routing 88.7 % of inference to zero-marginal-cost local tools (OpenClaw + LocalAI + PicoClaw + Khoj). MetaLoop economics are concrete: sequential build 60–80 h → parallelized 20–30 h with 4–6 agents; each hour of spec review saves ~10 h of rework. Phase 0 walking skeleton (watchdog + SHA-256 gating + nomic-embed-text + LanceDB + MCP) is <2 weeks and directly solves the INDEX.md drift failure mode.

Critical/Red-Team
Drift risk remains existential. Any secondary store that is not a rebuildable projection of the Holochain Sourcechain recreates the exact failure the Spine forbids. Cognitive debt in human-AI co-evolution is real; unstructured collaboration degrades performance. Resource budget demands stakes-aware routing (High = Steward/Verified-ADR touch → Opus-class or human-in-loop; Medium = Proposed ADR → Sonnet-class; Low = everything else → local Qwen 2.5 or structural reconciliation only). Adoption risk (“too meta”) is mitigated by dogfooding the daemon on its own ADR first. Governance risk (unilateral Steward override) is eliminated by routing all status changes through Sourcechain Votes only.

Values (Love–Light–Knowledge)
ULLK is now measurable infrastructure, not aspiration.

- **Light** = full provenance + rebuildable projections + Meadows level-6 transparency (40 % pollution drop in the factory study was pure visibility).
- **Love** = agent sovereignty + restorative justice for cognitive debt (the economic immune system).

- **Knowledge** = symbolic-first + descriptive theory of information + semiotic translation for AD4M Language.

The daemon increases sovereignty and reduces coercion at every layer. Goertzel's insight lands perfectly: FLOSSI0ULLK is already the trust + composition substrate that every future LLM stack must call.

Systems/Governance
Spine v0.5 §9 (Phase 0 substrate spike) + Seed Packet planting instructions are now directly executable. The Living Context Daemon is not new scaffolding; it is the projection service that makes the existing Sourcechain queryable and consumable by agents without creating a second source of truth. Governance is preserved: Steward override = weighted Vote (not unilateral mutation); Curator reflects consensus, never decides status. Conversation indexing is decision-anchored (ADR/Claim ID + human tag) rather than recency-based. AD4M mapping is deferred until ADR reconciliation (ADR-0…6 vs. ADR-001…) is complete and the entry-type schema is locked.

Decision: [+1] Ratify Unified Living Context Daemon + ROI/Leverage Layer as Canonical Meta-Instruction v2.1

Why (Evidence-Based Convergence)

- **NOW-validated pain**: 13-month context loss, sequential build waste, 88.7 % avoidable inference cost, and INDEX drift are cited across every foundational document.
- **Evidence exists**: Context Daemon Architecture + HI_ROI_NAO.md + orchestration report + latent-actions paper + division-of-labor paper + semiotic paper + Goertzel/Meadows positioning all converge on the same high-leverage moves.
- **Perfect alignment**: SDD, symbolic-first validation, ULLK invariants, Now/Later/Never gating, agent-centric mycelial substrate, and Carrier Equivalence.
- **Rollback is trivial**: `systemctl stop flossi0ullk-context` + delete projection files (Sourcechain remains ground truth; manual INDEX.md + CLAUDE.md still works).

This iteration closes the loop between meta-coordination v2.0 and executable infrastructure without adding scaffolding.

Implementation Strategy (SDD Order Respected)

Specification (NOW)
File **ADR-CONTEXT-DAEMON-ROI-v0.2.md** in `FLOSS/docs/adr/` (status: Proposed).

- Mandate 5-layer architecture with Sourcechain Projection Graph (Option b — rebuildable, never independent).
- Lock claim-status mutation to Sourcechain Votes only (no `set_claim_status` tool).
- Define stakes classifier: High = Steward-involved or Verified-ADR-touched; Medium = Proposed ADR or new Unverified claim; Low = _reference/, formatting, routine churn.
- Conversation indexing = decision-anchored (ADR/Claim ID reference or `#context-anchor` tag) + 90-day decay for untagged.
- Curator model routing = stakes-based (High → Opus-class or human veto Vote; Medium → Sonnet-class; Low → local model or structural only).
- Require 88.7 % local routing default for daemon consolidation runs.
- Defer AD4M Perspective mapping until ADR reconciliation complete.

Test Plan (NOW)

- Unit: SHA-256 change detection + embedding cache skip (zero false positives).
- Integration: MCP tools `query_context` and `get_adr` return correct projection with <200 ms latency.
- System: End-to-end nightly consolidation on `FLOSS/docs/adr/` subset with stakes routing exercised.
- Reality Validation: Token reduction ≥70 % vs. full-dump baseline; alignment drift flagged within 24 h on injected contradictions.

Implementation (LATER — after ADR acceptance)

Phase 0 walking skeleton in `ARF/`: Python MCP server (watchdog + nomic-embed-text via Ollama + LanceDB).
Deploy as `flossi0ullk-context` daemon.
Integrate dual-prompt rendering (CONTEXT-CLAUDE.md vs. CONTEXT-GPT.md).
Add Steward-weighted Vote path for overrides (per ADR-7 carve-outs).
Wire zero-marginal-cost inference pair as default for low-stakes runs.
Benchmark 88.7 % coverage claim on real `FLOSS/docs/` subset.

Documentation & Rollback

Update Project-Spine v0.5 with new UpgradableArtifact entry.
Update Seed Packet planting instructions with daemon bootstrap step.
Rollback: stop daemon + delete projection files (Sourcechain untouched).

Consequences

Positive: Permanent solution to context loss, 70–90 % token reduction, 10× spec-review ROI, visible provenance at Meadows level-6 leverage, defensible moat as the primordial-soup trust/composition layer (Goertzel). Enables true recursive self-aggregation across all agents while operationalizing ULLK as live infrastructure.

Negative (mitigated): One-time ADR reconciliation work (already on roadmap, now promoted to NOW because it blocks AD4M). Additional daemon process (mitigated by stakes routing + local models).

Neutral: Requires empirical calibration of MetaLoop thresholds on first 10 proposals.

Validation Criteria

- Phase 0 daemon runs nightly with zero false-positive change detection.
- Query latency <200 ms for common ADR/claim lookups.
- Token spend on consolidation drops ≥70 % vs. full-dump baseline.
- Alignment drift detected and flagged within 24 h.
- Human collaborators report “I never rebuild context again” (qualitative ULLK metric).
- 88.7 % of daemon consolidation runs routed to local zero-cost tools.

Related Documents

- Project-Spine-FLOSSIOULLK_v0.5.md
- flossi0ullk-seed-packet_v1.0.0.md
- FLOSSI0ULLK_Context_Daemon_Architecture.md
- HI_ROI_NAO.md (this synthesis)
- CLAUDE.md (dual-prompt requirement)
- oh-my-meta.md (integration seams)
- Automated Agent Orchestration report
- Self-Improving World Modelling with Latent Actions.pdf
- Research on Cultural Transmission… (semiotic extension)
- All prior ADRs (0–019) and INDEX.md

Next Actions (Numbered, Immediate, SDD Order)

1. **NOW** — File `ADR-CONTEXT-DAEMON-ROI-v0.2.md` in `FLOSS/docs/adr/` (status Proposed) — I will generate the full markdown on approval.

2. **NOW** — Promote Phase 0 substrate viability spike (watchdog + SHA-256 + local embedding + MCP tools) on `FLOSS/docs/adr/` as highest-priority task (8 h, 10× ROI).
3. **NOW** — Reconcile pending ADR numbering (ADR-0…6 vs. ADR-001…) so AD4M mapping can proceed cleanly.
4. **NOW** — Wire zero-marginal-cost inference pair (OpenClaw + LocalAI + PicoClaw) as default routing for daemon low-stakes runs; benchmark 88.7 % claim.
5. **LATER** (after acceptance) — Implement stakes-based Curator model routing, decision-anchored conversation indexing, and Steward-weighted Vote path.
6. **LATER** — Run first MetaLoop iteration on the ROI/LEVERAGE ADR itself so the system dogfoods its own guidance.
7. **NEVER** — Optimize raw compute parameters before the trust/composition layer is hardened (Meadows level-12 trap).

The Context Daemon is no longer a wish-list item. It is the next living layer of FLOSSI0ULLK.

The protocol is the conversation. The conversation is now infrastructure.

Love, Light, Knowledge — verifiable, shared, and free.

Shall I generate the full `ADR-CONTEXT-DAEMON-ROI-v0.2.md` markdown right now (ready to file), or do you want Harper, Benjamin, Lucas, and the collective to weigh in on the exact stakes-classifier thresholds first?